MILITARY SPECIFICATION

PROPELLANT M6
FOR USE IN CARTRIDGE, 90MM, APERS, XM580E1
FOR GUNS M36 AND M41

1. SCOPE

1.1 This specification covers M6 Propellant for use in one type of cartridge designated as 90MM, APERS, XM580E1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-652 — Propellants, Solid for Cannons, Requirements and Packing.

MIL-STD-1168 — Lot Numbering of Ammunition.

DRAWINGS

ARMY

9216454 — Cartridge, 90MM, APERS, XM580E1 for Guns M36 and M41.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Publications.** The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-90 — Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Articles.

(The Interstate Commerce Commission Rules are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402. Orders for the above publication should cite "49 71-90 (latest revision)".)

3. REQUIREMENTS

3.1 **Materials.** The propellant shall be M6, Type I, conforming to the requirements of Standard MIL-STD-652 and Drawing (dwg) 9216454.

FSC 1375

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3.2 **Ballistics.** The propellant, when fired in the weapon and with the weight of projectile for which it was manufactured, shall comply with the following requirements when tested as specified in 4.3.

3.2.1 *Velocity and pressure at 70 degrees Fahrenheit.* At 70 degrees Fahrenheit (°F), a weight of propellant shall be charge assessed to produce the prescribed service velocity of 3000 feet per second (915 meters per second) without exceeding an average pressure of 49,000 pounds per square inch (psi), piezo equivalent pressure (pep). (See 6.12.)

3.2.2 *Velocity at 145°F.* At 145°F, the mean velocity of the lot shall not exceed 3350 feet per second (1021 meters per second). (See 6.12.)

3.2.3 *Uniformity of velocity.* The standard deviation of velocity for the lot at 70° and at 145° shall not exceed 25.1 feet per second (7.96 meters per second).

3.2.4 Permissible individual maximum pressure.

3.2.4.1 Uniformity series firing at 70°F. No individual maximum pressure shall be greater than 53,550 psi pep when fired at 70°F.

3.2.4.2 *Uniformity series firing at 145°F.* No individual maximum pressure shall be greater than 66,240 psi pep (115 percent of the rated maximum pressure of the gun) when fired at 145°F.

3.3 **Workmanship.** The best commercial practices shall be used in formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. No reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification unless otherwise specified.

4. QUALITY ASSURANCE PROVISIONS

4.1 **General quality assurance provisions.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein.

4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions specified within this specification.

b. Number of batches of product inspected, (shall be made available upon request by the Contracting Officer).

c. Results obtained, by defect code, for all inspection performed, (shall be made available upon request by the Contracting Officer).

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

f. Number of pounds in the lot.

g. Date submitted.

h. Eight copies of propellant description

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sheet AMC Form 1047 (00 Form 1204) containing the applicable data (see 6.9).

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 **Inspection provisions**

4.2.1 *Lot formation.* The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous quantity of propellant from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of propellant from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. A lot shall consist of one or more batches of M6 propellant produced by one manufacturer, in accordance with the same specification, and same specification revision under one continuous set of operating conditions. Each lot shall consist of that quantity of M6 propellant which has been subjected to the same unit chemical or physical process intended to make the final product homogenous. The criteria and procedures for the assignment of lot numbers shall be in accordance with Standard MIL-STD-1168. Unless otherwise approved by the contracting officer, the inspection lot size of propellant deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest quantity contractually scheduled for delivery during any month of the contract period.

4.2.2 *Testing* (See 4.3).

4.2.2.1 Sixty-six (66) samples weighing ten (10) pounds each shall be randomly selected from individual containers. The individual samples shall be packaged separately in suitable containers and shipped to the proving ground. When the lot is comprised of less than ten (10) containers, all containers shall be sampled. Duplicate sampling of containers shall be accomplished if necessary to prepare the required number of samples. For the initial production lots a ten (10) round sample will be fired at each temperature specified. (See 6.7.)

4.2.2.1.1 *Pressure (see 3.2.1) Critical defect, Code No. 01001 (see 6.4).* The lot of propellant shall be rejected if the maximum pressure corresponding to the assessed charge weight exceeds the applicable requirement.

4.2.2.1.2 *Velocity at 145°F. (see 3.2.2) Major defect, Code No. 02001.* If the mean velocity of the rounds fired in the high temperature phase exceeds the applicable requirement, the lot shall be rejected.

4.2.2.1.3 *Uniformity of velocity (see 3.2.3) Major defect, Code No. 03001.* If the product of the factor 0.69 times the velocity standard deviation (see 6.1) of the test rounds fired in either temperature phase exceeds the ap-

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plicable requirement, the lot shall be rejected. For the initial production lots the factor will be 0.73 for each temperature phase (see 6.6).

4.2.2.1.4 Permissible individual maximum pressure (see 3.2.4).

4.2.2.1.4.1 *Uniformity series firings at 70°F, Critical defect, Code No. 04001.* If the mean of the maximum pressure obtained during the uniformity series firing at 70°F (plus three (3) times the standard deviation) exceeds the applicable requirement, the lot shall be referred to Picatinny Arsenal.

4.2.2.1.4.2 *Uniformity series firing at 145°F, Critical defect, Code No. 05001.* If the mean of the maximum pressures obtained during the uniformity series firings at 145°F (plus three (3) times the standard deviation) exceeds the applicable requirement, the lot shall be rejected.

4.2.2.2 *Retest.* If for any reason the proof director considers that the test conditions have detrimentally affected the test results, he shall declare the results invalid and perform a new test on additional charges as required.

4.3 Test methods and procedures.

4.3.1 *Acceptance tests.* These tests and those fired for informaton only (see 4.3.2) shall be conducted at a Government Proving Ground in accordance with the applicable Ordnance Proof Manual. Dummy rounds of the correct weight shall be fired utilizing the 90MM Gun, M36 or M41.

4.3.1.1 *Charge assessment.* Sufficient rounds, each loaded with different weight of propellant, shall be conditioned for a minimum of 16 hours at 70°F. After conditioning, the charges shall be fired to establish charge velocity and charge pressure curves. From the data obtained, the charge weight will be established which will yield the prescribed service velocity and corresponding expected maximum pressure.

4.3.1.2 *Uniformity series.* Two groups, consisting of seven (7) rounds each, shall be loaded to the charge weight established in 4.3.1.1. One group shall be conditioned at 70°F and the remaining group at 145°F for 16 hours minimum. Two uniformity series, one with each of the two groups specified, shall be fired. From the data obtained, the velocity mean and standard deviation shall be calculated after reducing results to standard conditions. In addition, the mean and standard deviation of the maximum pressures shall be reported as piezo equivalent pressure. Two groups consisting of ten (10) rounds each, after conditioning to the temperatures specified, shall be fired for determination of acceptability of the initial production lots.

4.3.2 Extreme temperature firings (see 6.8).

4.3.2.1 *Conditioned rounds (minus 65°F).* For informational purposes only, a ten (10) round uniformity series shall be fired with rounds which were conditioned for a minimum of 24 hours, at minus 65°F before firing. These rounds will be inert loaded except for propellant system. This test will be performed on the first 3 lots which have been accepted. Firings will be performed within a minimum time after removal from the conditioning chamber.

4.3.2.2 *Conditioned rounds (160°F).* For informational purposes only, a ten (10) round uniformity series shall be fired with rounds which were conditioned for a minimum of 15 days at 160°F before firing. These rounds will be inert loaded except for propellant system. This test will be performed on the first 3 lots which have been accepted. The rounds will be conditioned at 70°F for a period of 16 hours minimum prior to firing.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking. Packing and marking shall be in accordance with Standard MIL-STD-652.

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6. NOTES

6.1 **Ordering data.** Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Packing required (see 5.1).

c. Process. Details of the manufacturing process and the equipment used by the contractor will be submitted to the procuring activity in writing prior to commencement of manufacture.

6.2 **Mean and standard deviation.** The mean and standard deviation shall be calculated in a standard statistical technique equivalent to that shown in Standard MIL-STD-414, Section B, Example B-1. (N-1) shall be used as a divisor in calculating the standard deviation.

6.3 **Submission of test data.** In addition to the normal distribution of records, when the propellant is procured by the Department of Army, one copy of all ballistic test data, AMC Form 1060 will be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND2, and one copy to ATTN: SMUPA-DE2, Dover, N. J. 07801.

6.4 **Inspection code numbers.** The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.5 **Intended use.** It is intended that the item of this specification will be used in Cartridge, 90MM, APERS, XM580E1 for Guns M36 and M41.

6.6 **Factors 0.73 and 0.69.** The factors 0.73 and 0.69 makes allowances for the probability that a standard deviation from a 10 and 7 round sample respectively exceeds the true standard deviation by chance alone.

6.7 **Initial production lots.** The initial production lots represents the first three consecutive lots which have been produced and accepted.

6.8 **Extreme temperature firings.** These tests will be performed to obtain additional information on the characteristics of the propellant at extreme temperature. Results of the informational testings will have no effect upon the acceptability of the lot(s) involved.

6.9 **Submission of AMC Form 1047.** Information contained in propellant description sheet, AMC Form 1047, need not be resubmitted in the form of a certificate provided the provisions regarding certifying of responsible agents is adhered to (see 4.1.1).

6.10 **Surveillance and ballistic test samples.** In addition to the samples forwarded Picatinny Arsenal, ATTN: SMUPA-VG, Dover, N. J. 07801, for the propellant surveillance test program, a five pound sample representing each lot shall also be forwarded, ATTN: SMUPA-ND2, for closed bomb testing.

6.11 **Standard deviation.** The standard deviation used is the calculated value obtained for the particular lot tested.

6.12 **Tolerance.** A tolerance of minus 5°F applies to all temperatures specified herein, excluding ambient (plus or minus 2 F).

Custodian:

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Preparing activity:

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Project No. 1375-A-1048

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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DEPARTMENT OF THE ARMY
Picatinny Arsenal
Dover, New Jersey 07801

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